WEEK 04 : PRODUCT CONTRIBUTION PHASE : DAYS 16 TO 21

Productionization, Microservices & Capstone

FastAPI Production Deployment, Docker Containerization & Executive Leadership Defense

PROGRAM & ROLE	COMPANY PRODUCT	TIME COMMITMENT	SCHEDULE & FEEDBACK
ML Engineering Intern (All-Round)	**Attenomy Cortex (Enterprise AI)**	**15-20 Hours/Week (Part-Time Unpaid)**	**Submit by Friday (11:59 PM) | Weekend Defense**

01 Production Engineering & Capstone Mandate

CAPSTONE SCOPE

In Week 4, you elevate your machine learning contributions to **Attenomy Cortex** from research experiments into an enterprise-grade, containerized production microservice. Your service will deploy with asynchronous FastAPI endpoints, Pydantic v2 request validation, Docker containerization, sub-50ms inference latency, an interactive demonstration dashboard, and a formal technical defense before the Attenomy executive engineering leadership.

02 Capstone Engineering & Automated Testing

DAYS 16-19

```python
# FastAPI production microservice with Pydantic v2 schema validation
from fastapi import FastAPI, HTTPException
from pydantic import BaseModel, Field
class InferenceRequest(BaseModel):
    query_text: str = Field(..., min_length=3, max_length=1000)
    context_id: str = Field(..., pattern=r'^[a-zA-Z0-9_-](8, 32)$')
    confidence_threshold: float = Field(0.75, ge=0.0, le=1.0)
@app.post('/predict', response_model=InferenceResponse)
async def predict(payload: InferenceRequest):
    return cortex_service.predict(payload)
```

TASK 4.1 End-to-End Capstone Project Engineering

Days 16-19 : Mon-Thu

Execute the complete end-to-end capstone project for Attenomy Cortex on the mentor-approved dataset, adhering to strict reproducibility standards and backing data pipelines with a comprehensive unit test suite in CI.

- **6 Core Capstone Components:** (1) Business problem statement tied to Attenomy Cortex metrics (sub-50ms latency, high precision), (2) Robust data ingestion and schema validation pipeline, (3) Condensed EDA summary, (4) Cross-validated comparison of baseline vs classical vs deep learning models, (5) Interpretability via SHAP / feature attributions, and (6) Ethical analysis covering adversarial drift and regional bias.
- **Unit Test Suite:** Implement at least 5 unit tests in **`tests/test_pipeline.py`** with `pytest` testing data cleaning edge cases, feature transformation shapes, model inference outputs, and threshold fallbacks.
- **CI Automation:** Configure a GitHub Actions workflow in `.github/workflows/ml-ci.yml` that installs dependencies, executes ruff linting, and runs pytest on every push.

ACCEPTANCE CRITERIA

✓ Repository runs end-to-end from README instructions in under 30 minutes of setup on a fresh machine.
✓ Final deployed model measurably beats baseline with documented cross-validation confidence intervals.
✓ At least 5 unit tests pass automatically in GitHub Actions CI with zero build warnings.

TASK 4.2 Asynchronous Microservice API & Interactive Demonstration

Days 19-20 : Thu-Fri

Deploy the model behind an asynchronous FastAPI microservice with input validation guards, containerize with Docker, and build an interactive demonstration dashboard in Streamlit or Next.js.

- **FastAPI Architecture:** Implement production endpoints in **`src/api/main.py`**: `POST /predict` (Pydantic v2 validation, pre-warmed model inference, latency header), `GET /health` (system status, model version hash, uptime), and `GET /metrics` (Prometheus request counts and latency percentiles).
- **Input Validation Guards:** Implement graceful exception handlers. Invalid payloads or out-of-range feature values must return explicit HTTP 422 JSON errors rather than server crashes.
- **Production Dockerfile:** Author a multi-stage Dockerfile based on `python:3.10-slim`, enforcing non-root user execution (`USER attenomy`), optimized layer caching, and single-command startup: `docker build -t attenomy-cortex-service . && docker run -p 8000:8000 attenomy-cortex-service`.
- **Interactive UI & Demo Video:** Deploy a Streamlit app (**`streamlit_app.py`**) enabling real-time token/query inspection and visual feature attribution. Record a concise 2-3 minute technical video walkthrough demonstrating setup, curl requests, and dashboard interaction.

ACCEPTANCE CRITERIA

✓ Valid input returns correct inference in under 50ms; invalid input returns clear HTTP 422 JSON errors.
✓ Service loads the saved pre-warmed pipeline in memory at startup (zero retraining on request).
✓ Service builds and executes inside Docker container with zero permission or dependency conflicts.

TASK 4.3 Executive Final Report, Model Card & Leadership Defense

Days 20-21 : Fri

Publish a formal 6-10 page technical report, create a 1-page standardized Attenomy Model Card, and defend the capstone system before the Attenomy engineering and leadership team.

- **Technical Report:** Publish **`reports/final_report.pdf`** (6-10 pages) structured into: Executive Summary, Business Framing, Data Architecture, Baseline & Advanced Models, Ablation Studies, Production Latency, Ethical Considerations, and Future Engineering Roadmap.
- **Standardized Model Card:** Deliver a 1-page document (**`reports/model_card.pdf`**) defining intended domain use, out-of-scope applications, training data distributions, primary performance metrics, ethical safeguards, and model versioning hash.
- **Executive Presentation:** Deliver a 12-15 minute slide deck (8-12 slides) with 5 minutes of technical Q&A to the Attenomy leadership team, including at least one slide on lessons learned from unexpected failures.

ACCEPTANCE CRITERIA

✓ A non-technical manager can understand the core problem, solution, and business impact from reading the first page.
✓ Presentation addresses edge-case failures, surprising empirical findings, and proposed mitigation architectures.
✓ Intern answers technical questions accurately and transparently regarding methodology and limitations.

ATTENOMY ENGINEERING STANDARD : PRE-WARMED MICROSERVICE INITIALIZATION

Production microservices must never fit or reload model weights per request. Always warm up model pipelines during FastAPI lifespan startup events: **@app.on_event('startup')** to guarantee deterministic sub-50ms inference latencies from the very first incoming ping.

03 Comprehensive 100-Point Evaluation Rubric

GRADING STANDARDS

Evaluation scale: 4 = Exceeds Expectations | 3 = Meets Expectations | 2 = Partially Meets | 1 = Does Not Meet. Total Score = Sum (Criterion Rating / 4 * Weight) * 100.

CRITERION	WEIGHT	MEETS EXPECTATIONS (3)	EXCEEDS EXPECTATIONS (4)
1. Data Handling & EDA	12%	Correct cleaning, clear EDA with useful insights	Thorough cleaning, hypotheses tested, risks anticipated
2. Modeling Depth	18%	Sound models, tuned, sensible comparison	Strong baselines, tuned gradient boosting, thoughtful comparison
3. Evaluation Rigor	15%	Correct split, 5-fold CV, metrics, no leakage	Proper CV, uncertainty reported, subgroup checks
4. Code Quality	15%	Clean, mostly modular, reproducible with README	Modular, tested, linted, runs with one command
5. Deep Learning	10%	Working model, curves plotted & interpreted	Diagnoses training behavior, effective transfer learning
6. Deployment & Eng.	8%	Working API/app with Pydantic validation	Robust FastAPI service, Docker, sub-50ms latency
7. Communication	12%	Clear and organized written report & deck	Compelling, tailored to executives, honest about limits
8. Professionalism	10%	Reliable, meets deadlines, applies feedback	Proactive, seeks feedback, consistent learning log

04 Performance Bands & Post-Internship Outcomes

CAREER PATHWAY

85-100 POINTS : OUTSTANDING
Strong candidate for immediate extension, return offer, or leadership on production client deployments.

70-84 POINTS : SUCCESSFUL
Met all program engineering goals; qualified for guided production tasks within Attenomy engineering.

55-69 POINTS : SATISFACTORY
Completed core requirements with identifiable gaps; targeted engineering improvement plan required.

BELOW 55 POINTS : DID NOT MEET
Did not satisfy core engineering objectives; recommended repeating foundational curriculum.

05 Actionable Daily Execution Checklist

INTERN CHECKLIST

- ☐ **Day 16:** Finalize capstone data pipeline, execute cross-validated model comparisons.
- ☐ **Day 17:** Implement 5+ unit tests in tests/test_pipeline.py, verify CI passes on GitHub.
- ☐ **Day 18:** Build FastAPI service in src/api/main.py with Pydantic v2 validation guards.
- ☐ **Day 19:** Author Dockerfile, build container, test local curl requests, build Streamlit UI.
- ☐ **Day 20:** Record 2-3m video, compile report & model card, submit by end of Friday (11:59 PM).
- ☐ **Day 21:** Present 15-minute capstone defense during weekend review, complete off-boarding.

06 Weekly Submission Deadline & Weekend Capstone Defense

FINAL GATE

Weekly Submission Deadline: There is no fixed daily stand-up. Interns work flexibly and must submit all FastAPI service code, Dockerfiles, Locust benchmark reports, and Capstone slide decks to the repository by end of Friday (11:59 PM).
Weekend Defense Schedule: On Saturday or Sunday, you will present your 15-minute presentation, 5-minute live microservice demonstration, and 10-minute technical interrogation during your scheduled 1-on-1 session with senior leadership.
Off-Boarding Checklist: Ensure all code is pushed to `main`, Docker images tagged in the container registry, and Model Card committed to `reports/model_card.pdf`.